

STAGE 2

Setting paths:

```
TARGET="/home/user01/ghidra-codeql-analysis/ghidra-10.0.4-  
public/ghidra_10.0.4_PUBLIC"
```

```
R="/home/user01/ghidra-codeql-analysis/reports"
```

```
LOG4J_JAR=$(find "$TARGET" -name "log4j-core-2.12.1.jar" | head -n 1)
```

```
JNDI_CLASS="org/apache/logging/log4j/core/lookup/JndiLookup.class"
```

```
echo "$LOG4J_JAR" | tee "$R/stage2-ghidra-log4j-jar-path.txt"
```

Manual inspection:

```
jar tf "$LOG4J_JAR" \
```

```
| grep -F "$JNDI_CLASS" \
```

```
| tee "$R/stage2-ghidra-jndilookup-jar-inspection.txt"
```

Syft,trivy,grype for JndiLookup:

```
grep -i "JndiLookup" "$R/syft-ghidra-cyclonedx.json" \
```

```
| tee "$R/stage2-ghidra-syft-jndilookup-search.txt"
```

```
grep -i "JndiLookup" "$R/trivy-ghidra-cyclonedx.json" \
```

```
| tee "$R/stage2-ghidra-trivy-jndilookup-search.txt"
```

```
grep -i "JndiLookup" "$R/grype-on-syft-sbom.json" \
```

```
| tee "$R/stage2-ghidra-grype-jndilookup-search.txt"
```

Locate jndilookup in Ggrype metadata:

```
jq -r '  
  paths(scalars) as $p  
  | (getpath($p) | tostring) as $v  
  | select($v | test("JndiLookup"; "i"))  
  | "\($p | map(tostring) | join(".")) = \($v)"  
' "$R/grype-on-syft-sbom.json" \  
| tee "$R/stage2-ghidra-grype-jndilookup-all-locations.txt"
```

Cdxgen class/namespace evidence:

```
jq -r '  
  .components[]? as $c  
  | ($c.properties // [])[]?  
  | select((.value // "" | tostring) |  
contains("org.apache.logging.log4j.core.lookup.JndiLookup"))  
  | "Component: \($c.name // "unknown") \($c.version // "unknown")"  
  + "\nPURL: \($c.purl // "none")"  
  + "\nProperty: \(.name // "unknown")"  
  + "\nClass: org.apache.logging.log4j.core.lookup.JndiLookup"  
' "$R/cdxgen-ghidra-jar-deep-cyclonedx.json" \  
| tee "$R/stage2-ghidra-cdxgen-jndilookup-confirmed.txt"
```